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Option: Software engineering (GL)

Development of an Adaptive Learning System with an AI Agent

Realized by:

Hammouche Mohamed Rached

Dammene Debbih Radhia

Sabri Chaker Iskander

Under supervision of:

Pr.Meriem Belguidoum

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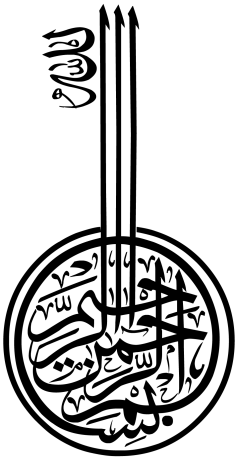
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General Introduction

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Part I

State of the Art

Chapter 1

Existing Systems Study

Introduction to Adaptive Learning Systems

An adaptive learning system (ALS) is a digital platform that personalizes the learning journey based on an individual's needs, progress, and level. It uses algorithms to analyze quiz results, identify strengths and weaknesses, and recommend where to focus next.

The importance of ALS lies in the fact that learners vary in terms of level, pace, and preferences. By adapting to these differences, ALS can improve learning outcomes and encourage long-term engagement.

1.1 What Are Skills?

1.2 Some Existing Adaptive Learning Systems

In this section, we explore a set of existing ALSs and examine their features to identify opportunities for innovation.

1.2.1 International Systems

- **Academy:**
  Khan Academy is a free online learning platform founded in 2008 by Salman Khan . It offers video lessons, adaptive practice exercises, progress tracking, and a teacher dashboard.
  It initially focused on mathematics but later expanded to other subjects such as science, computing, economics, history, and arts.
  The platform structures lessons in a hierarchical format divided into courses, units, lessons, and skills.

It recently introduced an AI-powered assistant called "Khanmigo," designed to support personalized learning and tutoring.

- **Points of strength:** Completely free access, clear and structured progression, trusted internationally, teacher monitoring tools.
- **Points of weakness:** Limited advanced specialization content compared to universities, adaptation mainly based on practice performance, limited gamification features.

- **Duolingo:**
  is a language-learning platform that uses a tree-structured skill progression system . It strongly focuses on gamification[1], offering various types of practice, including listening, speaking, vocabulary, and grammar exercises.
  It adapts to learners based on incorrect answers, repetition patterns, and response speed. It provides progress tracking, leaderboards, badges, levels, and streak-based motivation systems.
  The platform integrates artificial intelligence to predict weak skills and schedule personalized review sessions.

  - **Points of strength:** Highly interactive, strongly gamified, adaptive repetition system, mobile accessibility, long-term motivation mechanisms.
  - **Points of weakness:** Full features require subscription, limited to language learning, limited teacher control, limited deep conceptual explanations.

- **Coursera:**
  Coursera is a global online learning platform founded in 2012. It partners with well-known universities such as Stanford and Yale, as well as major companies such as Google, IBM, and Meta, to offer both paid and free courses, primarily video-based.
  It provides professional certificates, specializations, and degree programs, covering a wide variety of academic and professional domains.
  Although Coursera includes quizzes and assignments for assessment, its courses generally follow a linear structure and are not deeply adaptive in real time.

  - **Points of strength:** Academic credibility, strong university partnerships, scalable infrastructure, wide range of specializations.
  - **Points of weakness:** Many courses require payment, limited real-time adaptation, mostly linear progression, identical course paths for all learners.

[1]Gamification refers to transforming a formal process into a game-like experience, such as turning structured learning into an interactive game environment.

- **ALEKS:**
  ALEKS (Assessment and Learning in Knowledge Spaces) is an adaptive learning platform primarily used in mathematics and science education. It is based on knowledge space theory and uses artificial intelligence to assess student understanding precisely.
  The system evaluates students through diagnostic assessments and continuously adapts by selecting problems that target knowledge gaps. It is widely used in schools and universities.
  - **Points of strength:** Strong mathematical adaptation model, precise knowledge gap detection, widely adopted in formal education.
  - **Points of weakness:** Subscription-based access, limited subject diversity, less emphasis on gamification and engagement.

- **Sparrow:**
  Smart Sparrow was an adaptive e-learning platform that allowed educators to create interactive and adaptive courseware. It focused on higher education and professional training environments.
  The platform enabled instructors to design personalized learning paths based on learner responses and performance analytics.
  It was later acquired by Pearson[6] and integrated into its digital learning solutions then the original stand-alone platform was **discontinued**.
  - **Points of strength:** Instructor-driven adaptation design, advanced analytics, higher education focus.
  - **Points of weakness:** Complex authoring process, institutional focus rather than individual learners, discontinued as an independent platform after acquisition.

1.2.2 Local systems

1. **Moodle:**
   Moodle is a global open-source Learning Management System (LMS) widely used by universities and schools including some Algerian universities.
   It allows instructors to upload learning materials such as PDFs and videos, create quizzes and assignments, manage grades, and organize course content in a structured format.
   The platform also provides communication tools, including forums, announcements, messaging, and collaborative activities, enabling interaction between teachers and students.

- **Points of strength:** Open-source nature allows high flexibility and customization, widely adopted worldwide, strong institutional credibility, extensive plugin ecosystem.
- **Points of weakness:** Not inherently adaptive, typically follows a linear course structure, limited built-in AI assistance, interface can be less intuitive compared to modern platforms.

1.3 Comparative study

1.3.1 Comparison Criteria

In order to evaluate the previously presented systems, a comparative study was conducted based on five main criteria that represent the essential characteristics of ALS.

- **Level of Adaptivity**
  This criterion evaluates whether the system dynamically adjusts learning content, difficulty, or learning paths based on learner performance. It examines the presence of real-time adaptation mechanisms, knowledge gap detection, and personalized progression models.
- **Assessment Intelligence**
  This criterion measures the sophistication of the evaluation process. It includes the use of diagnostic assessments, mastery tracking, adaptive question selection, and data-driven learner modeling techniques.
- **AI Integration**
  This criterion assesses the extent to which artificial intelligence enhances the learning experience. It considers AI-powered tutoring assistants, predictive analytics, automated feedback generation, and personalized content recommendations.
- **Feedback Quality**
  This criterion evaluates how effectively the system helps learners understand and correct their mistakes. It examines the presence of detailed explanations, weakness detection, and personalized reinforcement strategies.
- **Teacher Involvement and Control**
  This criterion analyzes the degree of control and monitoring available to instructors. It includes access to analytics dashboards, the ability to modify content or learning paths, and tools that allow educators to influence adaptation mechanisms.

1.3.2 Comparative analysis

System	Level of Adaptivity	Assessment Intelligence	AI Integration	Feedback Quality	Teacher Control
Khan Academy	Moderate (adaptive practice)	Moderate	AI assistant (Khanmigo)	Good explanations	Strong dashboard
Duolingo	Moderate (skill repetition)	Basic to Moderate	AI-based prediction	Immediate corrective feedback	Limited
Coursera	Low (mostly linear)	Basic quizzes	Limited AI features	Standard feedback	Moderate (instructor-led)
ALEKS	High (knowledge space model)	Advanced diagnostic engine	Algorithm-based adaptation	Targeted remediation	Institutional control
Moodle	Low (conditional only)	Basic quiz engine	Minimal built-in AI	Depends on instructor	Strong customization

Table 1.1: Comparison of Existing Systems Based on Defined Criteria

Conculsion

The comparison between all these learning systems demonstrates that some of them adopt the adaptive learning definition and modern learning solutions, but we conclude that there is a clear gap in the field of **Adaptive Learning Platforms** due to the nonexistence of systems that combine comprehensive learning, assessment adaptivity, AI assistance, rule-based guidance, and teacher involvement.

Chapter

2

Design and Development methodology

Part II

Contribution

Chapter 3

Preliminary Analysis and Design

3.1 Project Description

This project aims to develop an AI-assisted Adaptive Learning System (ALS) that personalizes the learning experience for students based on their performance, progress, and learning pace. The application will allow students to create accounts, access structured learning content, and complete quizzes.

The system will dynamically adjust the difficulty level of questions according to each student's level and suggest learning materials using a **rule-based algorithm**. Students will be able to track their progress, view performance analytics, and receive encouraging achievements.

In addition to adaptive assessment, the system integrates an AI assistant that supports students by providing guidance, explanations, and learning recommendations throughout their progression.

The initial scope excludes mobile applications (Android/iOS versions), payment integration for premium courses, offline access mode, certification generation, and voice interaction with the AI assistant.

3.2 Project Objectives

This project aims to teach practical skills that are not commonly taught in universities and are usually gained through experience, such as writing clean code and isolating and correcting errors. These skills are organized into courses as packages that are essential in a developer's life.

To acheive this goal it is essential to:

- Provide a user-friendly web platform that delivers personalized learning experiences according to each student's performance and progress.

- Implement a rule-based algorithm that adjusts quiz difficulty and learning recommendations based on students' results.
- Track student progress and performance across quizzes to identify strengths and weaknesses in different topics.
- Integrate AI assistance to provide explanations, guidance, and learning support during the study process.
- Motivate students through gamification elements such as streaks, leaderboards, and achievement indicators.
- Ensure secure user management and access control, allowing users to register and log in safely.
- Allow teachers to create and manage **courses** and quizzes, and adapt learning content based on students' ratings and performance.
- Provide administrators with tools to manage courses, topics, content, and user accounts.

3.2.1 Functional Requirements

ID	Description	Priority
FR1	The system shall allow visitors to view the course list and public course information.	High
FR2	The system shall allow users to create an account using email/password or Google authentication.	High
FR3	The system shall allow registered users to log in and log out securely.	High
FR4	The system shall allow students to edit profile information and delete their accounts.	Medium
FR5	The system shall allow students to enroll in courses and access their learning roadmap.	High
FR6	The system shall allow teachers to create, update, and delete complete courses.	High
FR7	The system shall allow teachers to create, update, and manage quizzes for their courses.	High
FR8	The system shall allow teachers to view student quiz results and course ratings.	Medium
FR9	The system shall allow administrators to manage courses, topics, content, and user accounts.	High
FR10	The system shall allow students to take quizzes in different topics.	High
FR11	The system shall evaluate student performance based on quiz results and activity tracking.	High
FR12	The system shall adjust quiz difficulty according to the student's performance level.	High
FR13	The system shall recommend learning materials based on the student's strengths, weaknesses, and learning history.	High
FR14	The system shall provide students with a personalized dashboard displaying progress, weak points, achievements, and streaks.	Medium
FR15	The system shall save unfinished learning activities so students can resume them later.	Medium
FR16	The system shall allow students to optionally select preferred learning days and receive reminder notifications if they miss them.	Low
FR17	The system shall provide an AI assistant that answers student questions, gives explanations, and suggests learning guidance.	Medium
FR18	The system shall provide a leaderboard to display student rankings based on performance.	Low
FR19	The system shall allow administrators to view platform statistics and overall usage data.	Medium

Table 3.1: Functional Requirements of the Adaptive Learning System

3.2.2 Non-Functional Requirements

ID	Description	Priority
NFR1	The system shall be fast and responsive during user interactions.	High
NFR2	The system shall handle multiple users simultaneously without significant performance degradation.	High
NFR3	The system shall provide secure authentication and authorization for all users.	High
NFR4	User data shall be stored securely and protected against unauthorized access.	High
NFR5	The user interface shall be clear, intuitive, and easy to navigate.	High
NFR6	The system shall be responsive and adapt to different screen sizes (desktop, tablet, and mobile browsers).	High
NFR7	The system shall be compatible with major web browsers (Chrome, Firefox, Edge, Safari).	Medium
NFR8	The system shall support multiple languages (Arabic, English, and French).	Medium
NFR9	The system architecture shall allow future integration with external learning tools and AI enhancements.	Medium
NFR10	The system shall ensure data consistency and reliability during user interactions.	High
NFR11	The system shall maintain availability with minimal downtime.	Medium

Table 3.2: Non-Functional Requirements of the Adaptive Learning System

3.3 Requirement Specification

This section defines what will be built.

3.3.1 Actor Identification

The system includes the following actors:

- **Visitor**: A non-registered user who can browse limited public content (such as the home page and system description). However, the visitor must sign up or log in to access the adaptive learning features.
- **Student**: The primary user of the application who interacts with the adaptive learning system. The student can access personalized content, complete learning activi-

ties, and track progress.

- **Teacher**: A user who is responsible for creating and organizing educational content, creating quizzes, and tracking student progress.
- **Administrator (Admin)**: A user with elevated privileges responsible for managing user accounts, modifying the course or the skill list, and monitoring overall system activity.

3.3.2 Use Case Identification

Use Case	Actor	Priority
Sign up	Guest	High
View public homepage	Guest	High
Browse domains (Guest)	Guest	High
View course contents	Guest	Medium
Login	Student / Admin / teacher	High
Enroll in domain	Student	High
Browse domains (Student)	Student	High
View profile	Student	Medium
Modify information	Student	High
Delete account	Student	Medium
View student homepage	Student	High
Consult dashboard	Student	High
Take quizzes in subdomains	Student	High
Chat with AI agent	Student	medium
Consult leaderboard	Student	Medium
Browse domains (Teacher)	Teacher	high
Create course	Teacher	High
Edit course	Teacher	medium
Delete course	Teacher	Medium
View course (owned)	Teacher	Medium
Create a quiz	Teacher	High
Consult students reviews	Teacher	low
Manage accounts	Admin	medium
Manage learning trees	Admin	High
View platform statistics	Admin	low
Provide AI assistance	AI Agent	medium

Table 3.3: System Use Cases with Assigned Priority

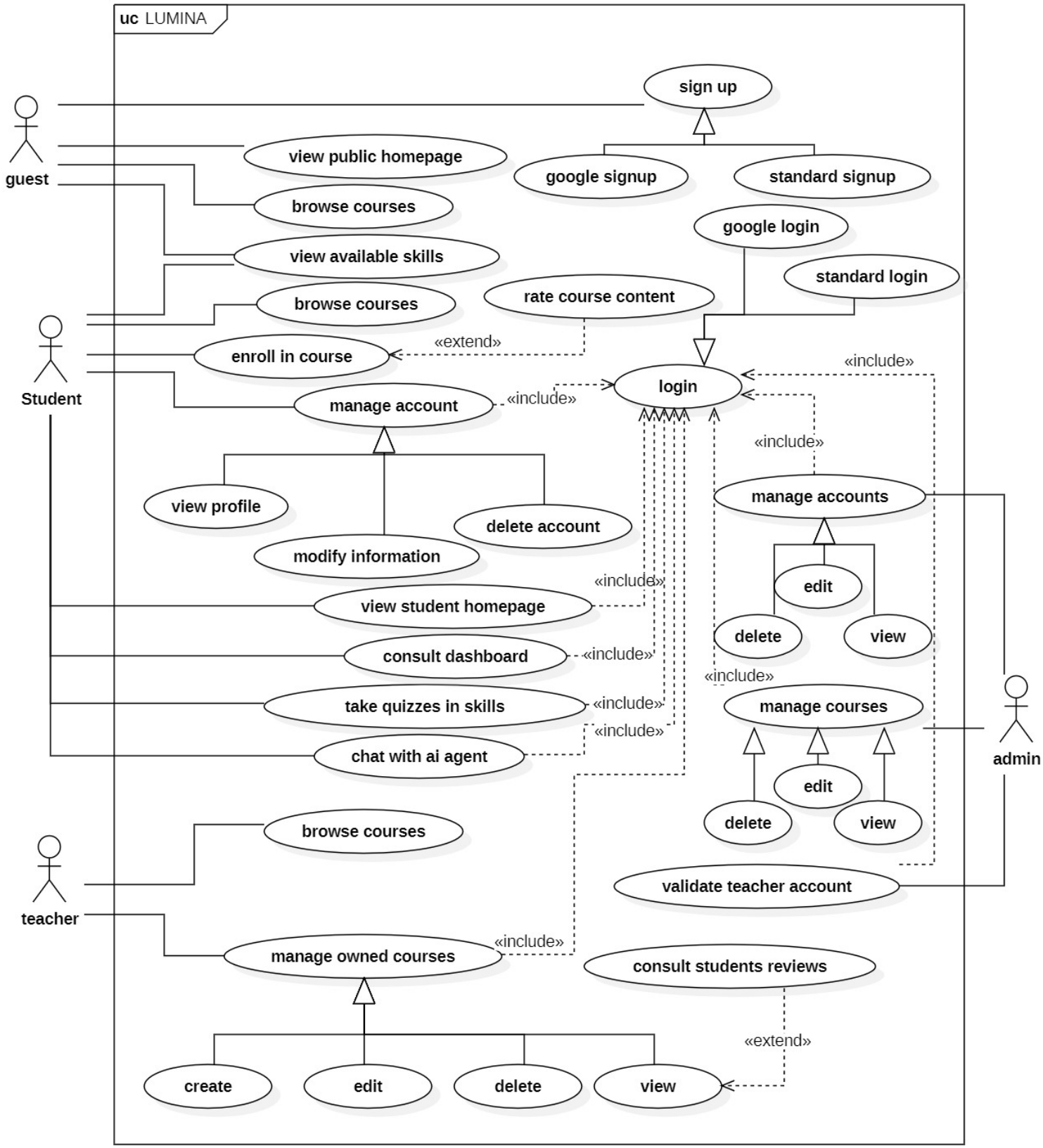

Figure 3.1: Use case diagram

3.3.3 User Stories

Guest user stories

User Story	Priority
As a guest, I want to browse available courses and skills so that I can choose what to learn.	medium
As a guest, I want to sign up as a teacher or a student, so that I can access the platforms' features	high

Table 3.4: Guest User Stories

Student user stories

User Story	Priority
As a student, I want to take quizzes to test my level.	high
As a student, I want to log in so that I can access my personalized dashboard.	high
As a student, I want to log out so that I can securely exit my account.	high
As a student, I want to see recommended domains so that I can focus on what suits my level.	medium
As a student, I want to see my uncompleted videos so that I can continue learning where I stopped.	medium
As a student, I want to see my streak so that I stay motivated.	low
As a student, I want to see my strong and weak points so that I know what to improve.	medium
As a student, I want to browse available domains so that I can choose what to learn.	high
As a student, I want to enroll in a domain so that I can start learning it.	high
As a student, I want to choose the type of learning material (videos, PDFs, or visual graphs) so that I can study in the way that suits my understanding.	medium
As a student, I want reminders to study so that I don't lose interests.	low
As a student, I want to see progress statistics so that I feel motivated.	medium
As a student, I want to manage my profile information.	high
As a student, I want to see a leader-board so that I know my position among others.	low
As a student, I want to rate a course so that teachers can improve it.	low
As a student, I want adaptive quizzes so that the questions match my skills.	medium
As a student, I want to chat with Ai assistant so that i can improve my knowledge.	medium

Table 3.5: Student User Stories

Teacher user stories

User Story	Priority
As a teacher, I want to upload my new course so that students can learn from it.	high
As a teacher, I want to update my course so that I can adapt its content.	high
As a teacher, I want to delete my course so that outdated content is removed.	high
As a teacher, I want to create quizzes for my course so that I can evaluate students.	medium
As a teacher, I want to update quiz questions so that assessments stay accurate.	medium
As a teacher, I want to view students' quiz results so that I can analyze their performance.	medium
As a teacher, I want to view students' ratings so that I can see what to adapt.	low
As a teacher, I want to view all available domains so that I can choose which domain to manage.	high
As a teacher, I want to log in so that I can manage my courses.	high
As a teacher, I want to log out so that I can securely exit my account.	high

Table 3.6: Teacher User Stories

Admin user stories

User Story	Priority
As an admin, I want to add a new domain so that students can learn new subjects.	high
As an admin, I want to update a domain so that its content stays accurate.	high
As an admin, I want to delete a domain so that outdated subjects are removed from the platform.	high
As an admin, I want to view all domains so that I can manage them easily.	high
As an admin, I want to create a course roadmap for each domain so that learning is structured.	high
As an admin, I want to update a course roadmap so that I can improve the learning path.	high
As an admin, I want to delete a course roadmap so that incorrect or outdated plans are removed.	high
As an admin, I want to view all student accounts so that I can monitor platform users.	high
As an admin, I want to deactivate or delete accounts so that I can manage inappropriate or inactive users.	high
As an admin, I want to view platform statistics so that I understand overall performance.	low

Table 3.7: Admin User Stories

3.4 Conception and Analysis

In this chapter, we present the conception and design of our application following the Agile Scrum methodology. Each sprint is detailed through its user stories, acceptance criteria, and UML diagrams including Participation class diagram, detailed sequence diagram, and activity diagram. This approach allows us to trace the evolution of the system incrementally, from planning to the final implemented features."

3.4.1 Domain class diagram

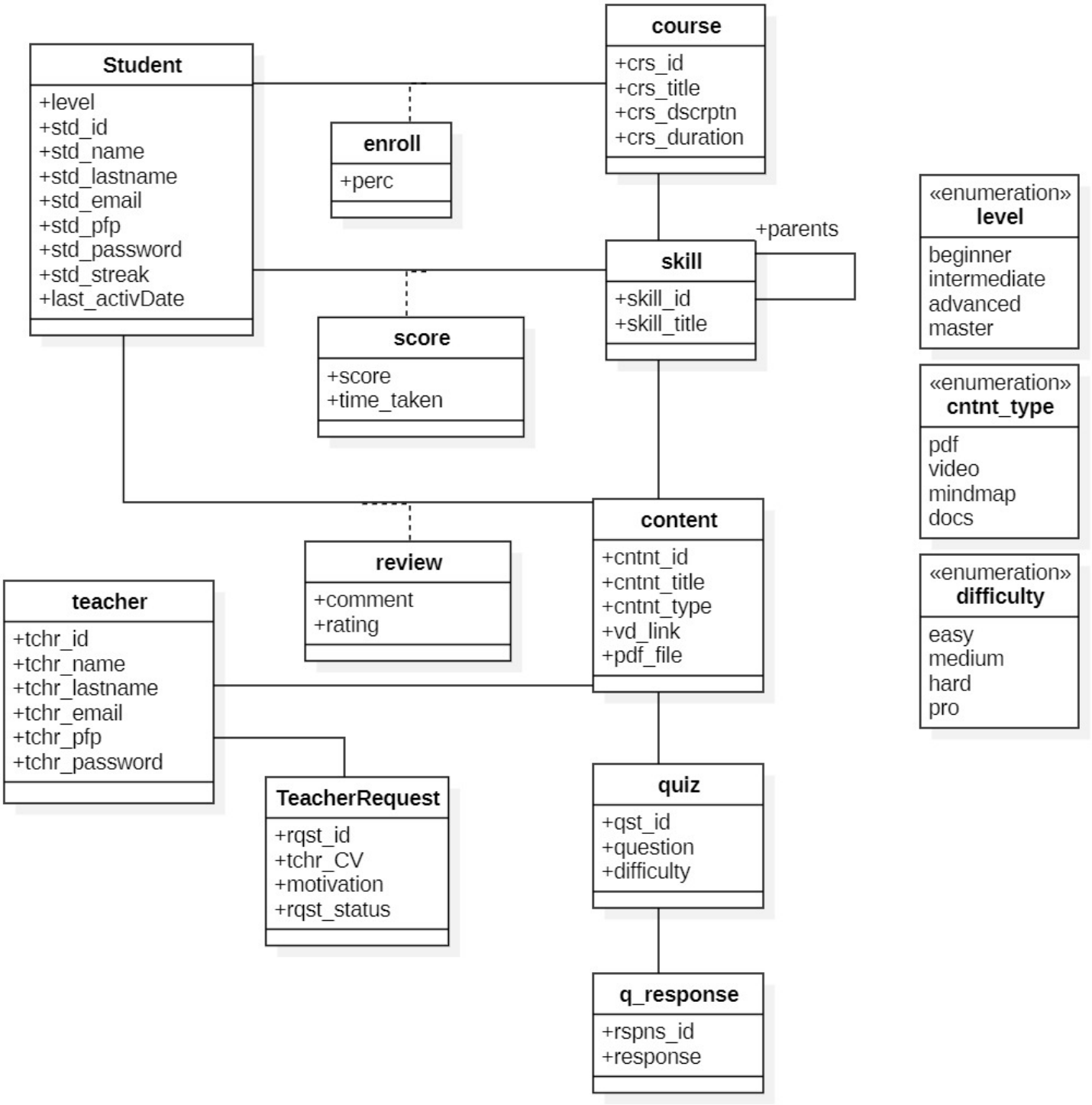

Figure 3.2: Domain class diagram

3.4.2 Sprint1 : Authentication

This first sprint covers the authentication module of the LUMINA platform. We address the two main use cases: user registration, which follows a differentiated flow based on the chosen role, and Admin validation. Each use case is detailed through its user story and corresponding UML diagrams.

ID	Task Name	Duration
1	**Sprint1: Authentication and authorisation**	**5 days**
2	Student register	1 day
3	Teacher register (2 step signup)	1 day
4	Student/teacher login	1 day
5	Admin login	1 day
6	Teacher account validation	1 day

Figure 3.3: Sprint1 organisation

User Registration:

1. **User Story :** As a guest, I want to register as a student or teacher, so that I can access the platform's features."
2. **Acceptance Criteria:**
   - Guest can choose between student and teacher role
   - Student completes 1 step and is logged in immediately
   - Teacher completes 2 steps including University, Specialty, CV upload
   - Teacher account is created with disabled features untill admin validates the account
   - A validation request is saved for the admin to review
   - Teacher sees a "pending approval" message after registration

3. **Participation Class Diagram :**

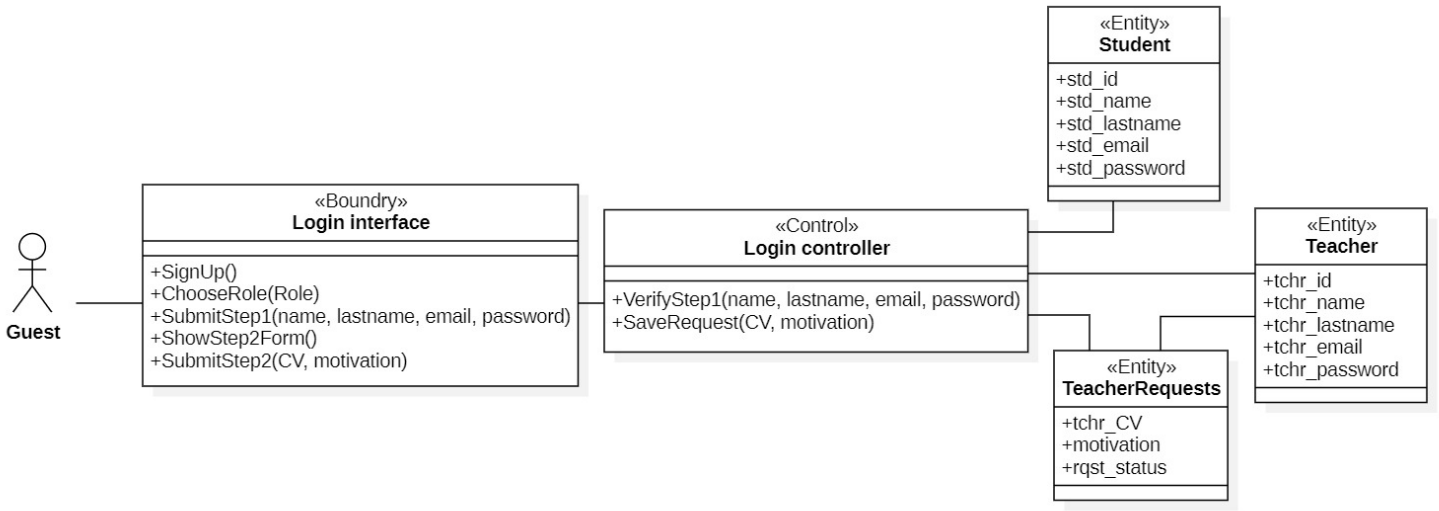

Figure 3.4: Participation class diagram for Sign Up Use case

4. **Detailed Sequence Diagram :**

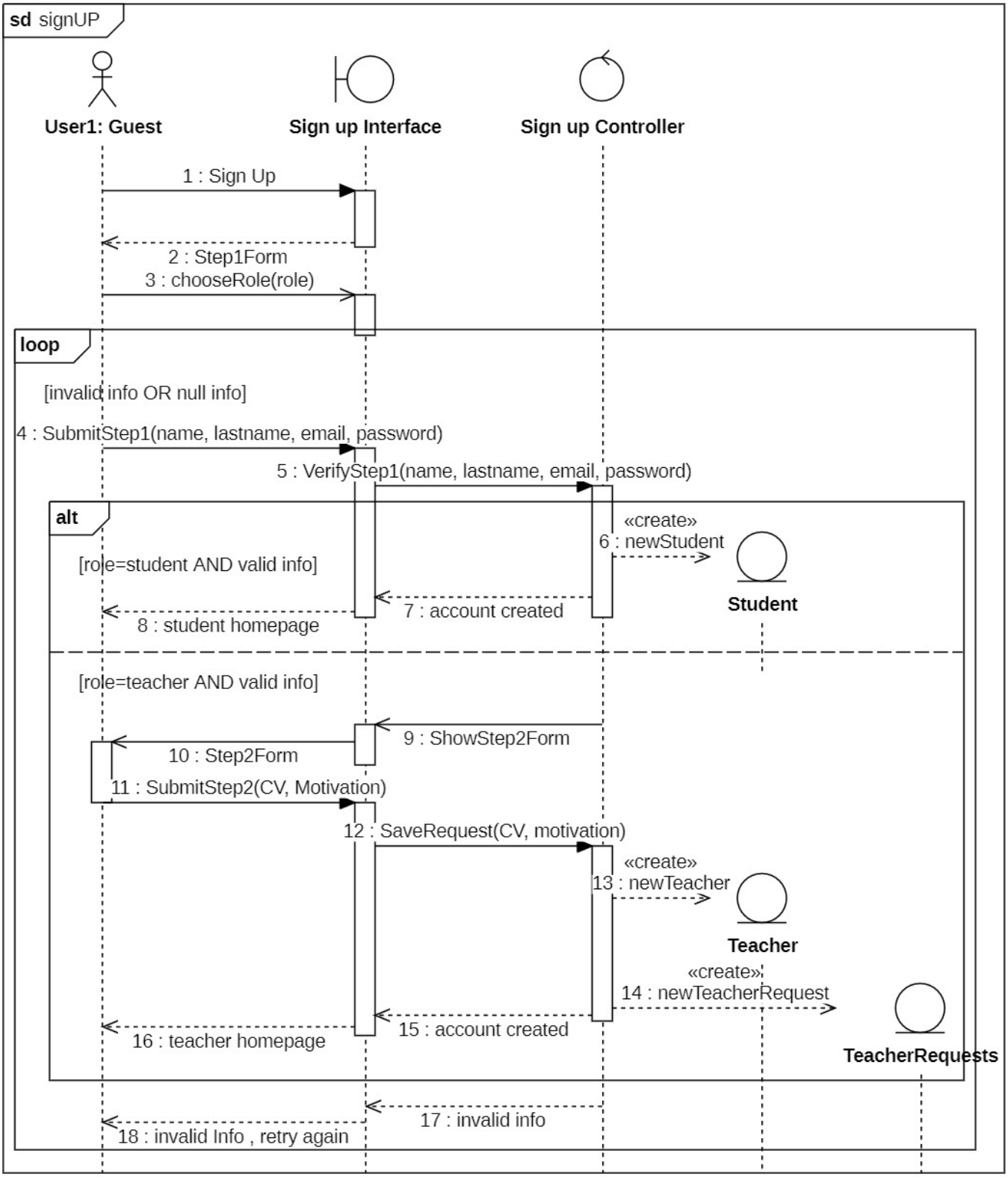

Figure 3.5: Detailed Sequence diagram for Sign Up Use case

Admin Validation:

1. **User Story :** As an admin, I want to review and validate pending teacher registration requests, so that I can ensure only verified teachers can access teaching features on the platform.
2. **Acceptance Criteria:**
   - Admin can view the list of all pending teacher validation requests.
   - Each request displays the teacher's basic info, uploaded CV and diploma.
   - Admin can approve a request → teacher account becomes fully activated.
   - Admin can reject a request → teacher account is marked as rejected.
   - Teacher receives a notification upon approval or rejection.
   - Approved teacher is updated to have "approved" status.
3. **Participation Class Diagram :**

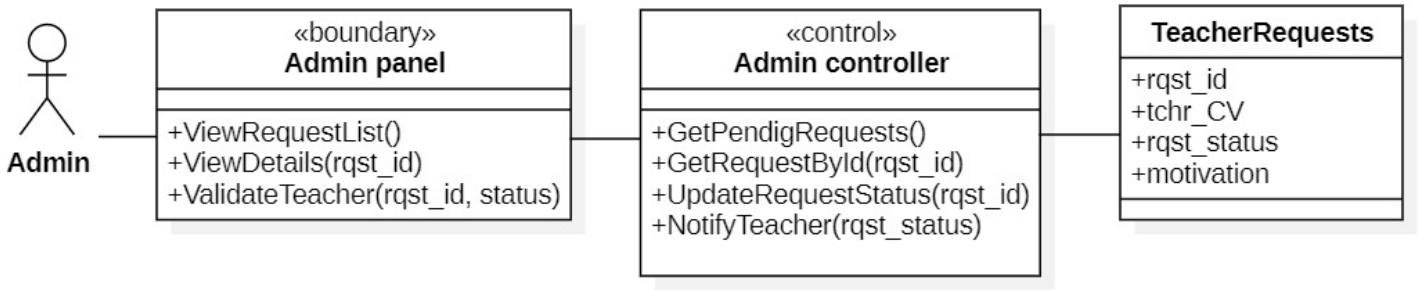

Figure 3.6: Participation class diagram for Validate Teacher Account Use case

4. **Detailed Sequence Diagram :**

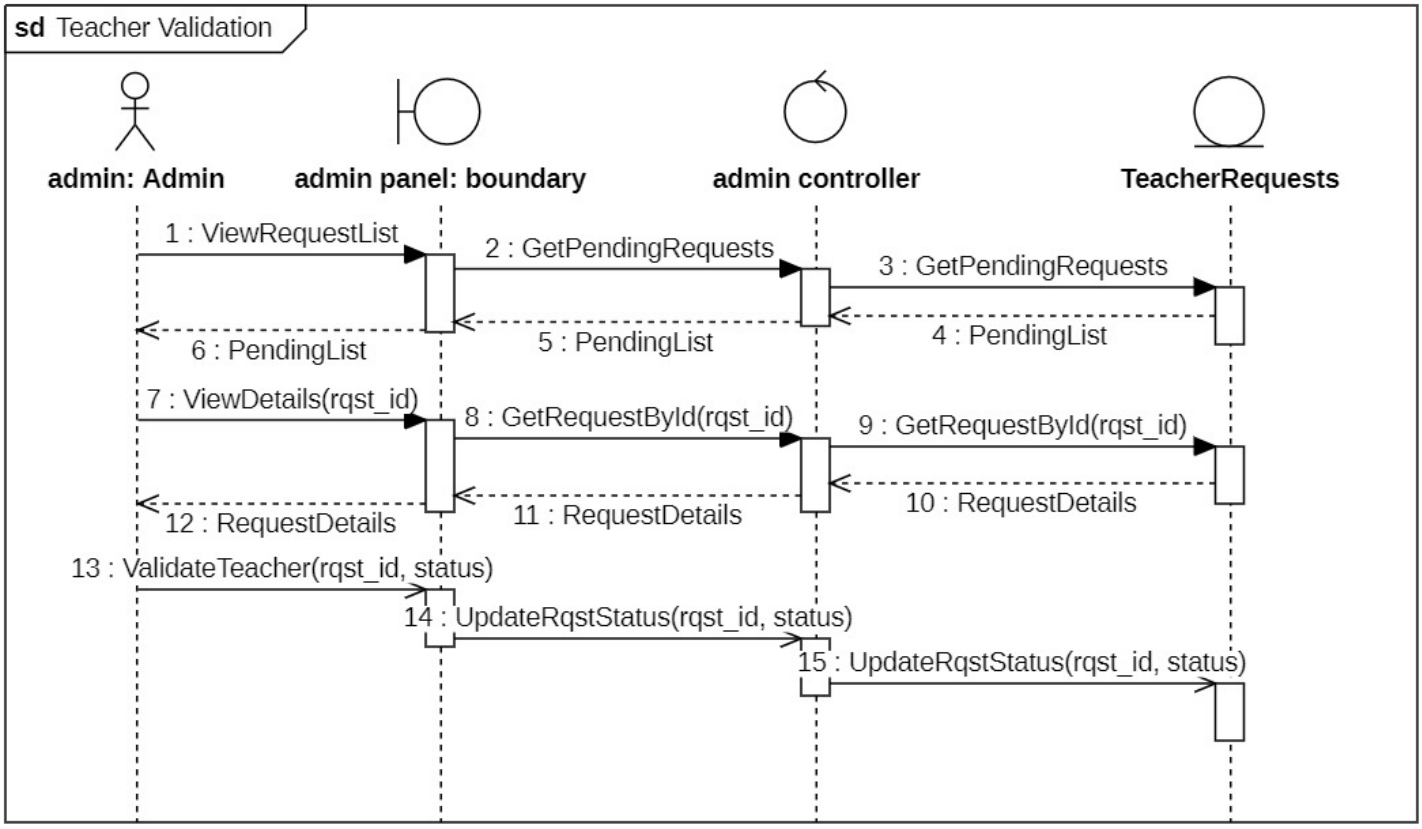

Figure 3.7: Detailed Sequence diagram for Validate Teacher Account Use case

Chapter

4

Implementation

Introduction

In this chapter, we present the implementation of our project. We describe the technologies and tools used to develop the Adaptive Learning System and explain the methodology followed during its development. We also present the organization of the team and how tasks were distributed to realize this work.

4.1 Technologies and tools

Category	Name	Definition / Purpose
Frontend Framework	NextJs	A React-based framework used to build the web application interface, handle routing, and support server-side rendering.
Frontend Library	React	A JavaScript library used to build reusable and interactive user interface components.
Styling Framework	Tailwind	A utility-first CSS framework used to design responsive and modern user interfaces.
Styling Language	CSS	Used to apply custom styles and layout adjustments.
Backend Platform	Supabase	An open-source backend service used for authentication and backend functionalities.
ORM Tool	Prisma	An Object-Relational Mapping tool used to manage and interact with the database in a structured way.
Database	PostgreSQL	A relational database system used to store application data.
Cloud Database Service	Neon	A cloud-based service used to host and manage the PostgreSQL database.
Deployment Platform	Vercel	A cloud platform used to deploy and host the web application.
Version Control	Git	A version control system used to track and manage code changes.
Collaboration Platform	GitHub	A platform used to host repositories and support team collaboration.
Development Environment	VS Code	A code editor used for writing and managing the project source code.
UI/UX Design Tool	Figma	A design tool used to create and prototype the user interface.
Project Management Tool	Jira	A project management and issue-tracking tool used to plan tasks, manage sprints, and monitor development progress.

Table 4.1: Technologies Used in the Project

The selected technologies were chosen for their scalability, reliability, and industry adoption, including modern frontend frameworks, backend services, database solutions, cloud hosting, collaboration, and project management tools

4.2 Sprint Organisation

Following the Scrum methodology, we organized the development of LUMINA into a series of sprints, each targeting a specific set of functionalities derived from the product backlog. The total development period spans approximately 12 weeks, divided into successive sprints of roughly one to two weeks each. Each sprint was planned to deliver a functional and testable increment of the application. Figure 4.1 below presents the complete sprint organization adopted throughout the project

ID	Task Name	Duration
1	**Sprint1: Authentication and authorisation**	**5 days**
2	Student register	1 day
3	Teacher register (2 step signup)	1 day
4	Student/teacher login	1 day
5	Admin login	1 day
6	Teacher account validation	1 day
7	**Sprint2 : Course Management**	**10 days**
8	Course creation	3 days
9	Create skill and define relation to course	2 days
10	Create topics and define relation to skill	1 day
11	Create a quiz for a topic	2 days
12	Consult course list	1 day
13	Course enrollement	1 day
14	**Sprint3 : Learning management**	**12 days**
15	Consult courses' skills and topics	1 day
16	Take quiz in a topic	3 days
17	Access learning content	3 days
18	Consult dashboard	3 days
19	Quiz evaluation and score counting	2 days
20	**Sprint 4 : Adaptivity and recommendation**	**8 days**
21	Weak/strong topics evaluation	5 days
22	Score-based topic locking/unlocking	3 days
23	**Sprint 5 : AI Integration**	**9 days**
24	AI assistant	6 days
25	AI recommendation	3 days
26	**Sprint 6 : Motivation features**	**4 days?**
27	Learning reminders	1 day?
28	Streak feature	1 day
29	Leaderboard Feature	1 day
30	Acheivements and global level feature	2 days

Figure 4.1: Sprint Organisation

4.3 Sprint 1 Implementation

In this section, we present the concrete realization of the authentication module developed during Sprint 1. We showcase the implemented interfaces for user registration and login, reflecting the flows defined in the design phase. Each interface is accompanied by a brief description of its functionality and behavior.

4.3.1 User interface mockups

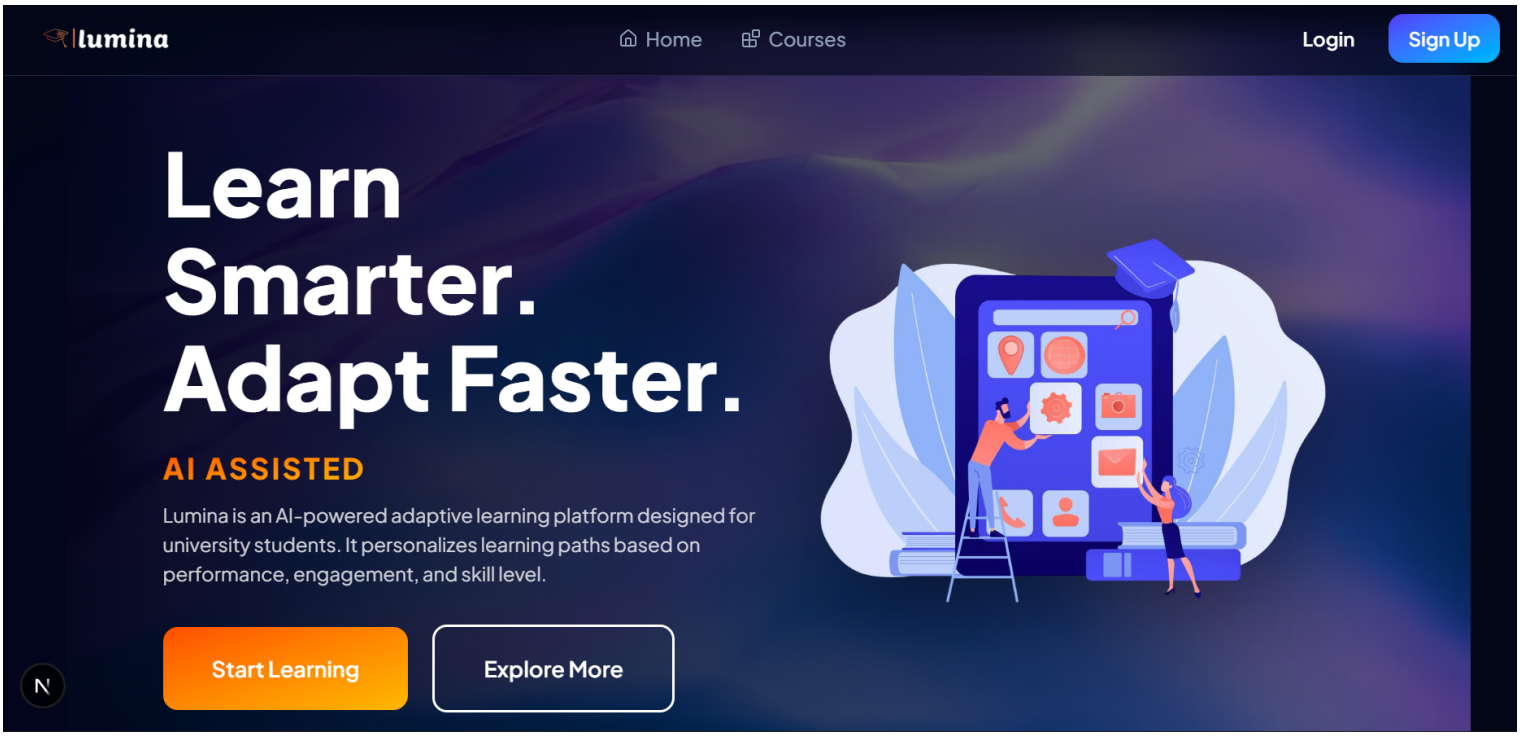

Figure 4.2: UI mockup for the hero page

Figure 4.2 illustrates the public homepage of the LUMINA platform, accessible to any guest visitor. The page presents the platform's value proposition along with two call-to-action buttons: Start Learning and Explore More. The navigation bar provides access to the Courses section, as well as the **Login** and **Sign Up** buttons, allowing guests to either authenticate or create a new account.

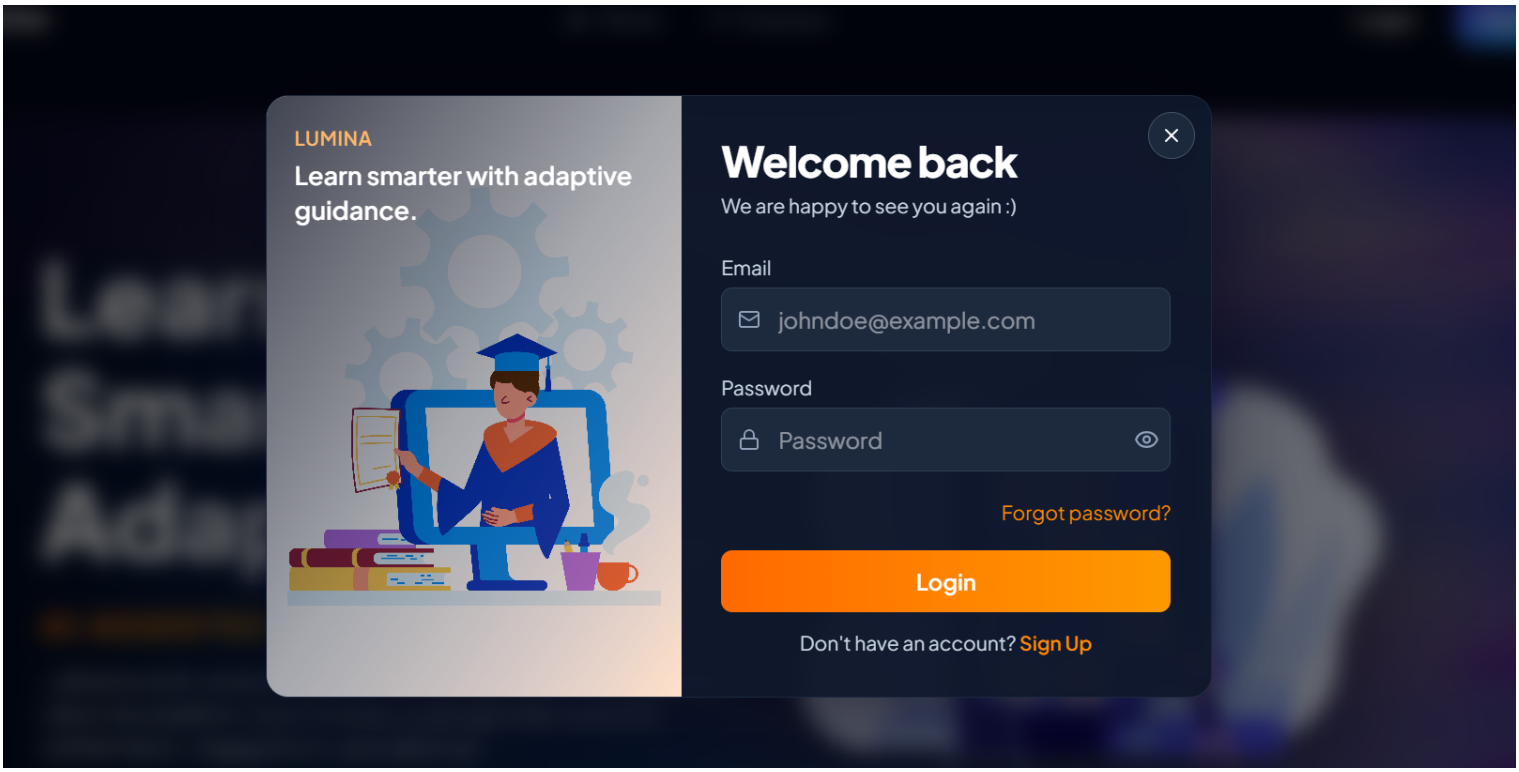

Figure 4.3: UI mockup for the login page

Figure 4.3 shows the login interface presented to returning users. The form requires an email address and password, with a Forgot password? link available for account recovery. A redirect link to the Sign Up page is also provided for users who do not yet have an account.

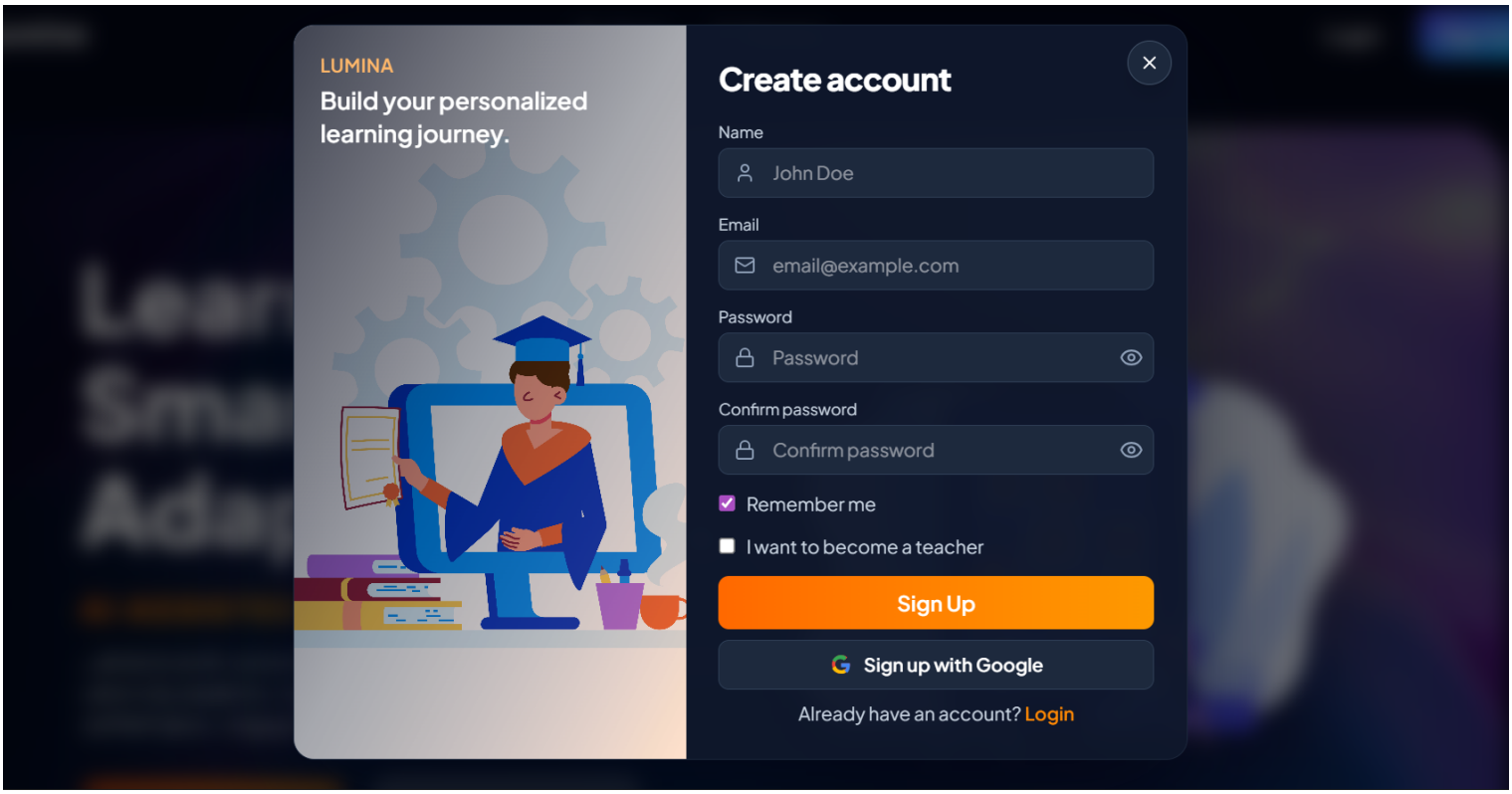

Figure 4.4: UI mockup for the sign up page

Figure 4.4 presents the registration form displayed upon clicking the Sign Up button. The guest is prompted to fill in their full name, email address, password, and password confirmation. An "I want to become a teacher" checkbox is available at the bottom of the form — if left unchecked, the user registers as a student and gains immediate access to student homepage upon submission. If checked, the user is redirected to a second registration step dedicated to teacher-specific information (CV and a motivation letter, Figure 4.5). Additionally, a Sign up with Google option is provided for faster registration

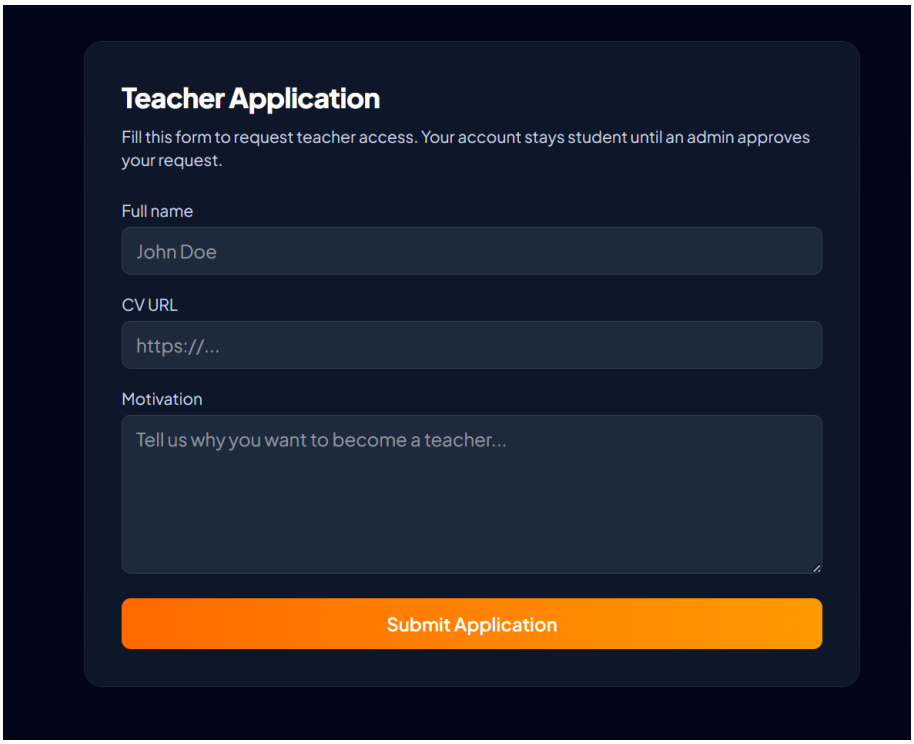

Figure 4.5: UI mockup for the Teacher sign up page

Figure 4.5 illustrates the second step of the registration process, exclusively triggered when the guest checks the 'I want to become a teacher' checkbox in Step 1. The applicant is required to provide their full name, a URL linking to their CV, and a motivation statement explaining their intent to teach on the platform. Upon clicking Submit Application, the system creates an account with restricted features (as shown in Figure 4.6)

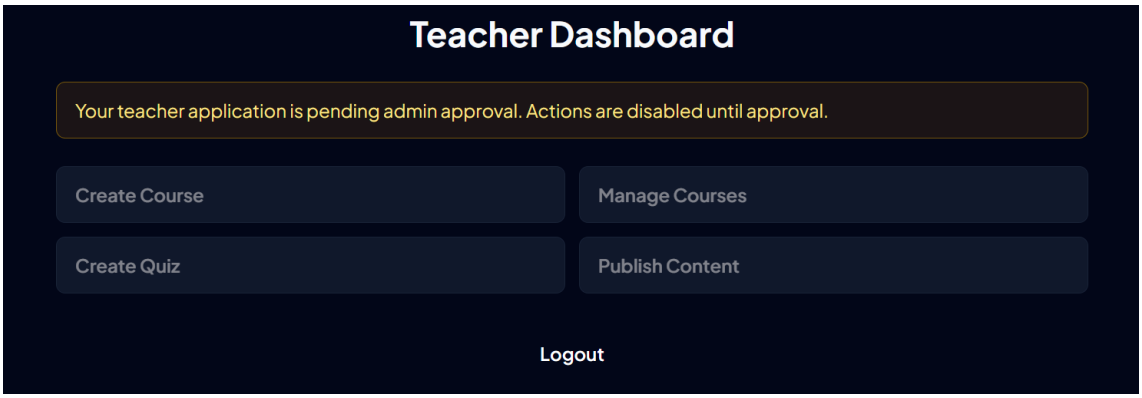

Figure 4.6: UI mockup for the disabled features Teacher page

4.3.2 Backend implementation

1. **Standard Login**

```
async function onSubmit(e: React.FormEvent) {
    e.preventDefault();
    setLoading(true);
    setError("");

    const { error: signInError } = await
       supabase.auth.signInWithPassword({
      email,
      password,
    });

    if (signInError) {
      setError("Invalid email or password. Please try again.");
      setLoading(false);
      return;
    }
    const {
      data: { user },
    } = await supabase.auth.getUser();

    const role = user?.app_metadata?.role;

    if (role === "student") router.replace("/student");
    else if (role === "teacher" || role === "teacher_pending")
       router.replace("/teacher");
    else if (role === "admin") router.replace("/admin");
    else router.replace("/");

    setLoading(false);
  }
```

The login function begins by preventing the default form submission behavior and initializing the loading state. It then calls the Supabase authentication service using the provided email and password credentials. If authentication fails, an error message is displayed and the process is stopped. Upon successful authentication, the system retrieves the current user's data and extracts their role from the application metadata. Based on this role, the user is redirected to the corresponding dashboard: a student is directed to the student interface, a teacher or pending teacher to the teacher interface, and an admin to the administration panel. This role-based redirection ensures that each user accesses only the interface appropriate to their profile.

2. **Standard Sign up**

```
async function onSubmit(e: React.FormEvent) {
    e.preventDefault();
    setError("");
    setSuccess("");

    if (password !== confirmPassword) {
      setError("Passwords do not match.");
      return;
    }

    if (password.length < 8) {
      setError("Password must be at least 8 characters.");
      return;
    }

    setLoading(true);

    const { data, error: signUpError } = await
       supabase.auth.signUp({
      email,
      password,
      options: {
        data: {
          full_name: fullName,
        },
      },
    });

    if (signUpError) {
      setError(signUpError.message);
      setLoading(false);
      return;
    }

    if (data.session && !rememberMe) {
      await supabase.auth.signOut({ scope: "global" });
    }
    if (data.session) {
  const res = await fetch("/auth/finalize-signup", {
    method: "POST",
    headers: { "content-type": "application/json" },
    body: JSON.stringify({ teacher }),
  });
```

```
  if (!res.ok) {
    setError("Account created, but setup failed.");
    setLoading(false);
    return;
  }

  const payload = (await res.json()) as { nextPath?: string };
  router.replace(payload.nextPath ?? "/student");
  router.refresh();
  setLoading(false);
  return;
}
    setLoading(false);
  }
```

The registration function starts by validating the input data on the frontend: it verifies that the two entered passwords match and that the password meets the minimum length requirement of 8 characters. If either condition fails, an error message is displayed and the process is halted before any server call is made. Once validation passes, the function calls the Supabase authentication service to create a new account using the provided email, password, and full name. If the registration fails, the returned error is displayed to the user. Upon successful account creation, the function checks the 'Remember Me' preference — if disabled, the session is immediately invalidated to prevent persistent login. A request is then sent to the internal endpoint `/auth/finalize-signup`, passing the user's chosen role (student or teacher). This endpoint is responsible for finalizing the account setup, assigning the appropriate role, and returning the redirection path. The user is then redirected to their corresponding interface, either the student dashboard or the teacher application page in the case of a teacher registration

3. **Sign up with google**

```
async function signUpWithGoogle() {
    setError("");
    setSuccess("");
    setGoogleLoading(true);
    const callbackUrl=new
       URL("/auth/callback",window.location.origin);
    if(teacher){
      callbackUrl.searchParams.set("wants_teacher","1");
    }

    const { error: googleError } = await
       supabase.auth.signInWithOAuth({
      provider: "google",
      options: {
        redirectTo: callbackUrl.toString(),
      },
    });

    if (googleError) {
      setError(googleError.message);
      setGoogleLoading(false);
    }
  }
```

The Google authentication function initiates an OAuth sign-in flow using the Supabase Google provider. Before triggering the redirect, the function constructs a dynamic callback URL pointing to the internal endpoint `/auth/callback`. If the user has checked the 'I want to become a teacher' checkbox, a query parameter `wants_teacher=1` is appended to the callback URL, allowing the server to detect the intended role after the Google authentication completes. The function then calls the Supabase OAuth service which redirects the user to Google's authentication page. Upon successful authentication, Google redirects the user back to the constructed callback URL, where the server finalizes the account setup and assigns the appropriate role. If the OAuth process fails at any point, an error message is displayed to the user.

General Conclusion

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Acronyms

(You can list the acronyms used in the document, for example:)

NTIC New Technologies of Information and Communication

UML Unified Modeling Language